

Linkability of LLM Digital Twins: Open-World Rates, a Twin-to-Twin Channel, and a Null That Absorbs Our Own Coupling

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Abstract

Published LLM “digital twins” — per-respondent simulated answer vectors released alongside survey panels — remain linkable to the humans they were built from. On the Twin-2K-500 panel (2,058 respondents), the strongest published twin designates the correct respondent in a closed pool of 2,058 in 23.23 % of cases [21.5 ; 25.0] under a rarity-weighted likelihood attack whose parameters are estimated out-of-fold, against 0.049 % for chance and under 0.3 % for every non-LLM predictor we tested at comparable marginal accuracy. On the Park et al. archive (1,052 participants) the same attack reaches **90.40 %** [88.6 ; 92.2] against 65.51 % for a naive agreement attack — our first attack understated that leakage badly. The rate is not an artifact of the closed pool: in an open world, at a false-accusation rate of 1 %, the attacker is right 4.28 % [3.3 ; 5.6] of the time on Twin-2K-500 and 60.17 % [54.5 ; 64.4] on the Park archive (against 3.04 % [2.0 ; 4.0] and 20.39 % [15.7 ; 24.5] for the naive attack — a gain that is statistically clear on Park and does not survive its interval on Twin), with the statistical baselines at roughly 0 % and a human test-retest ceiling of 54.5 % / 90.7 %. We report a twin-to-twin channel: within one pipeline, two twins of the same person built from the same persona source designate each other with no real human answer held by the attacker — 36.4 % top-1 at 60 common items on Twin-2K-500 and 11.9–13.0 % at 177 common items on the Park archive, against anti-artifact controls of 0.06 % and at most 0.31 %. Tested instead across genuinely independent pipelines, that threat model is refuted: top-1 falls to 1.8 % [0.4 ; 3.6] at $n = 142$, below a 9.2 % demographic baseline. We do not claim an absence of prior work: in a regime it does not evaluate, we contradict Anonymeter’s conclusion that linkability is the weakest of the three risks it measures — it links two attribute partitions of one real dataset with the attacker holding real values, where our attacker holds none. We also report what we could not establish. We predicted, and preregistered, that the observed coupling between imitation quality and leakage reflects an individual fingerprint; a null carrying only a matched per-person accuracy margin — an object that is not structure-free, and that in fact leaks more than our own twin (31.6 % top-1 against 20.73 %) — reaches a rank correlation at least as high as ours (0.974 [0.950 ; 0.993] against 0.965 observed). Fifteen preregistered predictions were refuted and two are inconclusive for lack of power; our own twins failed to reproduce the leakage even with a paid frontier model on the per-item recipe; and the twin-to-twin channel’s cross-dataset behaviour is

explained only in part, by item redundancy rather than item count. Following Das, Zhang and Tramèr, we built the control this literature was missing, and it absorbs our own effect. Finally, a 1998 mechanism (PRAM) applied to twins reduces closed-world top-1 from 20.68 % to 0.13 % at a measured cost of 4.4 points on inter-item correlations, and holds at 0.29 % against an adaptive attacker who knows the mechanism — while differential privacy, contrary to our own preregistered prediction, is not dominated by it on aggregate utility.

Keywords

digital twins, large language models, re-identification, linkability, synthetic survey data, privacy

1 Introduction

Survey research is beginning to publish *digital twins*: for each human respondent in a panel, an LLM is conditioned on that person’s profile and its simulated answers to the questionnaire are released as a per-person record. Toubia et al. published 2,058 such twins as Twin-2K-500 [38]. Park et al. built generative agents for 1,052 participants and cited privacy in restricting access to individual agent responses [27] — although the replication package that is publicly posted does carry the 1,052 participants’ individual responses together with those of the agent conditions we analyse (§ 4.1, § 8). The privacy question these releases raise has been named — by Park et al. themselves, by Bonagiri et al.’s normative call [4], and by the AAPOR task force [32], which ranks synthetic response generation as the most risky of the core tasks it evaluates and names re-identification by linkage without ever quantifying it — but it has not been measured.

This paper measures it, and reports both what the measurement supports and what it refutes.

1.1 What we set out to show, and what happened

Our central hypothesis was that a twin’s fidelity to *its particular person* and that twin’s identifiability are expressions of one underlying axis: make the simulation better and you necessarily make the person more findable. We preregistered a self-refutation test for this hypothesis. It has two parts. The first part passed: the rank correlation between imitation quality and leakage survives measuring the two axes on completely disjoint item sets (Spearman 0.969, 95 % CI [0.937 ; 0.993], 50 stratified random A/B splits of 60 items; 100 % of splits above 0.7) [c7-disjoint-resultsats.md § 1]. The relation is therefore not an artifact of recycled computation.

The second part failed, and it failed against us. We constructed a *marginal null*: 100 artificial predictors matched to each person’s own accuracy margin, with correct positions drawn at random

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without looking at the person — an object that is not structure-free, and that in fact leaks more than our own twin (31.6 % top-1 against 20.73 %). Adversarial review later found a real defect in that first witness, which has since been rebuilt without changing the verdict (§ 5.1). The rebuilt witness reaches mean Spearman **0.974**, 5th–95th percentiles [**0.950** ; **0.993**], against **0.965** observed over the 12 configurations [audit-renversement-2026-09-12.md]; the observed value does not exceed it. The 0.969 just quoted is that same observed quantity measured on the 50 disjoint A/B splits rather than on whole items; each null is compared only against the observed value measured on its own plan. **Reproducing the coupling requires nothing more than a matched per-person accuracy and correct positions drawn at random — prediction (b) is refuted.**

We report this as a result rather than a confession. A recent line of work argues that an attack on foundation models proves nothing until the null hypothesis is properly sampled: blind baselines routinely beat state-of-the-art membership inference attacks [8, 11], and a SaTML position paper makes the demand explicit [42]. The quality-leakage coupling reported in prose by Ward et al. [39] and Byun et al. [6], and contested by Platzner and Reutterer [29] and Adams et al. [1], has never been tested against such a control. We built the control that was missing, and it absorbs our own effect.

What survives this is not a theory of why twins leak. It is a set of five measurements that do not depend on the coupling being individual-specific at all.

1.2 Contributions

We order these from the least expected to the most.

(1) **A twin-to-twin linkage channel within a shared pipeline — and, tested separately, refuted once the pipeline is genuinely independent.** Two twins of the same person, sharing their persona source and differing only in model or output format, designate each other when they share enough items — while the attacker holds no real human answer of any kind. On Twin-2K-500: 36.4 % mean top-1 over 30 pairs sharing 60 common items, against an anti-artifact control (decoy from the same demographic segment) of 0.04 % on those pairs. On the Park archive: 11.9 % [10.1 ; 13.8] interview→survey and 13.0 % [11.2 ; 14.9] survey→interview at 177 common items, against a control of at most 0.31 %. We never report a single number for this channel without its common-item count (§ 5.4). Anonymeter’s contrary conclusion [18] is reached under a different definition — two records of the *same* real dataset, the attacker holding *real* attribute values — in a regime it did not test. Built end-to-end independently instead (T2, § 3), top-1 falls to **1.8 % [0.4 ; 3.6]** at $n = 142$, below a 9.2 % demographic baseline: the 36.4 % figure is not evidence for that scenario (§ 5.4).

(2) **The leakage is localised in the pattern of answers, not in memorised content.** An ablation destroys the signal by permuting the order of a twin’s answers while leaving their content untouched — 33.1 % to 0.046 % top-1 (§ 5.5): what identifies is the dependence structure between answers, not any answer’s content, which is what a regurgitation account would require. Structurally, the pipeline also trains nothing on the target population, so no train/test gap exists to exploit, unlike Yeom et al. [41] and Feldman [14]. We state

the limit of this plainly rather than let it be found: our training-cutoff control (llama31-8b, cutoff preceding the panel’s publication, copies nothing verbatim and leaks no more than models that do [c7-gen-resultats.md]) is a control on **our own** twins, and it lives in a regime where none of our twins identifies almost anyone (§ 5.7). It therefore cannot settle the question for the published twins that carry this paper’s rates, whose training data and cutoffs we cannot probe. **We do not claim a measured dissociation between memorization and re-identification.** The narrower claim we do make — a structural mechanism, localised by ablation — is what separates our channel from the memorization [7] and free-text inference literatures [23, 24, 35].

(3) **Open-world rates with their human ceiling.** Closed-world 1-in-N rates assume the target is in the pool; the defensible measurement drops that assumption and reports true detections against false accusations (§ 5.3, Figure 1).

(4) **A defense measured with its real cost, and held against an adaptive attacker.** Not a new mechanism — a variant of the 1998 Post Randomisation Method [20] — but applied to LLM twins with a risk-utility curve that is measured rather than assumed, reported with the cost component a single average would hide, tested against an attacker who knows how the defense works, and measured on analyses a practitioner would actually run (§ 6). Against our own preregistered prediction, differential privacy is *not* dominated by it on aggregate utility (§ 6.2).

(5) **An attack strong enough to falsify our own first measurement.** A rarity-weighted likelihood attack with out-of-fold parameters raises the Park archive’s closed-world top-1 from 65.51 % to 90.40 % and triples its open-world rate at 1 % false accusations — a correction of our own published figure (§ 5.3, § 5.8).

(6) **A bits-of-identity instrument that transports across datasets where the raw rate does not.** A derived instrument, not a discovery: a case of Rényi min-entropy leakage [34], in the spirit of Eckersley’s surprisal [13], addressing the transportability problem of the Rocher et al. scaling law [31] by another route (§ 5.6).

1.3 Negative results, stated in front

Fifteen preregistered predictions were refuted and two are inconclusive (Table 1, § 7.1). Our own twins identify almost nobody, and the constructive demonstration meant to close that gap — raise fidelity with a frontier model and watch the leakage return — **failed**: openai/gpt-4.1 on Twin’s per-item recipe left fidelity at 0.1714 against their 0.708 (§ 5.7). We still do not know what the Twin-2K-500 recipe does differently, though the unknown is narrower: neither the model nor the call granularity. And the coupling that motivated this work does not survive its own control (§ 5.1). We consider a reader who stops here to have read the paper honestly.

2 Background and Related Work

2.1 Privacy of synthetic data

Stadler, Oprisanu and Troncoso show that generated tabular data remains linkable to its sources [36]. Giomi et al. formalise three attacks — singling out, linkability, inference — and conclude, on their datasets, that linkability is the weakest risk [18]. That conclusion is contested: Annamalai, Gadotti and Rocher [2], Ganey and De Cristofaro [16, 17] and Ganey alone [15] show reconstruction

attacks defeating distance-to-closest-record (DCR) metrics; Yao et al. [40] and Meeus et al. [25] generalise the critique; Golob, Penttala and De Cock make these attacks an emerging standard [19].

What distinguishes us. This corpus concerns tabular microdata from classical generative models (GANs, copulas, diffusion), evaluated by continuous similarity metrics. Our object is a vector of categorical answers produced by an LLM conditioned on a person, where no similarity metric substitutes for a direct matching attack.

2.2 Re-identification of real data

Narayanan and Shmatikov re-identify Netflix users [26]; de Montjoye et al. show four spatio-temporal points suffice on credit-card metadata [9]; Rocher, Hendrickx and de Montjoye model risk on incomplete samples [30] and extend it into a scaling law [31]. Taub, Elliot, Pampaka and Smith ground the statistical measurement of disclosure risk on synthetic data (CAP, TCAP) on the premise that synthetic generation breaks the link between identity and datum [37].

What distinguishes us. These works re-identify from genuine auxiliary data. Our attack's input is never a real datum of the target: it is a model output generated from a persona, compared against real answers the attacker separately holds. We falsify precisely the CAP/TCAP premise, on LLM-simulated survey microdata.

2.3 Privacy and LLMs

Carlini et al. document memorization and regurgitation of training sequences [7]. Staab et al. infer personal attributes from innocuous free text [35]. Ko et al. de-anonymise online authors by multi-cue agentic inference [23] — the closest threatening neighbour — and Lermen et al. demonstrate the same attack at scale [24]. Theoretically, Yeom et al. connect membership attacks to overfitting [41]; Feldman shows memorizing rare cases is necessary for generalization [14]; tracing codes [5, 12] are the closest formal parent to our 1-in-N attack, bounding the worst case of an optimal adversary recovering an individual from aggregate statistics, though in dimension far exceeding the number of individuals.

What distinguishes us. Those works all start from free text bearing direct semantic cues, aggregated by an agent reasoning over it. Our channel is narrower and drier: closed-choice categorical answers, no free text, no agentic reasoning, and a mechanism localised in the dependence structure rather than in memorised content — established by ablation (§ 5.5), the training-cutoff control of § 5.7 bearing only on our own twins. Unlike Yeom and Feldman, our pipeline trains no model on the target population.

2.4 Respondent simulation and twins

Argyle et al. launched the use of LLMs as panel substitutes [3]. Park et al. cite privacy in restricting access to their 1,052 agents' individual responses [27], while the publicly posted replication package nonetheless carries individual participant and agent responses (§ 8) — a risk anticipated, never measured. Toubia et al. publish the 2,058-twin panel we attack, treating neither privacy nor linkage [38]. Peng et al. document group-fidelity distortions on that same panel without connection to re-identification [28]. Bonagiri et al. issue a normative call to evaluate these risks without measuring them [4]. None of these works, nor the five surveyed in our prior-art scan,

measures a re-identification rate from twin outputs. That is the gap we fill.

2.5 Closest prior art and contradictors

Ward et al. test 13 attacks over 9 generators and 48 tabular datasets, and state in prose a relation between quality and leakage close to ours [39]; Byun et al. observe a near-monotone frontier of the same type without a coefficient [6]. Three differences separate them from our claim: their quality is *aggregate* (MMD, Jensen-Shannon divergence, downstream AUC), never a correspondence to the right person; their risk is *membership* in a training set, not 1-in-N identification; and they publish no coefficient — the only figure in Synth-MIA is DCR against Max-AUC at $r = 0.225$, a weak correlation between two aggregate metrics.

Three contradictors hold that fidelity and risk separate: Platzer and Reutterer [29]; Adams et al., for whom a generator without formal guarantees preserves fidelity and utility without evident privacy harm [1]; and Shafieinejad et al., observing decoupling as quality saturates within a single tabular diffusion model trained longer [33]. Our response is not that they are wrong: all three measure *aggregate* fidelity, never individual correspondence to the right person, and none tests whether the coupling they observe survives a null matched on per-person accuracy. Ours does not (§ 5.1). Their decoupling is a different axis from ours: theirs unfolds over training time within one model (quality plateau against rising membership-inference risk); ours is a cross-sectional correlation across twelve heterogeneous methods against 1-in-N re-identification.

2.6 Defenses

Our D4 is not new. Shuffling answers between units of the same segment is a variant of the Post Randomisation Method [20], which perturbs categorical variables through a known transition matrix leaving margins invariant in expectation, and of data swapping, which preserves them exactly at high aggregations — the same family as the partially synthetic data tradition of Reiter and Drechsler, extended by Drechsler [10] and Bowen et al. [22]. What is new is the application to LLM twins with a measured risk-utility curve (§ 6).

3 Threat Models

We distinguish three scenarios. Only the first is the classical one, and we assume its strongest objection rather than deflecting it.

3.1 T1 — Panel holder publishes unkeyed twins; attacker holds the real answers

A panel holder releases per-person twin records without the key linking each twin to its respondent. An attacker who holds **the real answers to the same items** — an insider, a panel co-owner, a leak of the source file — matches twins to respondents. This is a *linkability* result in the sense of the Article 29 Working Party and the GDPR, not identification from public information: the quasi-identifier required is a specific questionnaire, rarely held outside the panel.

We state plainly what this is not. It does not show that a third party recovers a named person from public information. And on Twin-2K-500 itself the matching is trivial without any attack —

pid equals TWIN_ID, and the twin files copy StartDate, EndDate, Duration and RecordedDate line by line from the humans [c7-contre-examen-2026-09-11.md § 4]. The result therefore documents a generic risk for anyone releasing “anonymous” twins without that key, a practice Twin-2K-500 does not follow.

The objection “this is not real re-identification, the attacker already holds the true answers” is assumed, not rebutted. Our answers to it are three: the explicit model above, the open-world measurement of § 5.3 which gives the honest number, and T2, which removes the assumption entirely.

3.2 T2 — Two organisations publish twins of the same cohort; attacker holds nothing real

If several organisations each publish their own twins of the same panel with sufficient item overlap, a third party cross-references them holding no human answer at all. This threat model has no direct precedent in the synthetic-data literature: Anonymeter’s linkability presumes real attribute values in hand [18], and Guépin et al. remove the real-auxiliary-data assumption but target membership in a single generator’s training set, never matching between two independent generations [21].

Tested directly, and refuted. § 5.4’s 36.4 % twin-to-twin top-1 is a same-team, same-persona-file measurement — only the model or output format varies — and does not speak to T2 as stated. Built as T2 actually describes, two pipelines sharing nothing but the target person give top-1 **1.8 % [0.4 ; 3.6]** ($n = 142/200$, stopped by an OpenRouter transport error), below both the 9.2 % demographic baseline and the 25.5 % segment decoy, and lower than at $n = 82$ (3.7 %). T2 is not supported once the pipeline is genuinely independent; the 36.4 % figure is not evidence for it (§ 5.4).

3.3 T3 — Pre-publication self-audit

A panel holder runs the attack against their own file before publication, to decide whether a defense is required — the constructive use of the method, and the one the artifact (§ 9) serves.

3.4 Attacker capabilities, all models

The attacker performs no training on the target population, has no query access to the generating model, and uses only rank-based matching over categorical vectors. Costs are negligible (§ 9).

4 Data and Methods

4.1 Datasets

Twin-2K-500 [38], 2,058 respondents, public under CC BY 4.0. Target: wave 4, 60 items usable in common out of 108 (40 purchase items of the form “would you buy this product at this price?”, 20 opinion items). Attacker context where applicable: 494 context items from waves 1–3 plus 14 demographic variables. Chance top-1 in the closed pool is 0.049 %; chance top-10 is 0.49 %. A human test-retest condition (the same people answering waves 1–3 and wave 4) gives the empirical ceiling: 81.6 % [79.9 ; 83.2] top-1, accuracy 0.745.

Park et al. archive [27], 1,052 participants, 177 common items, replication package publicly posted. The archive documents five

agent conditions (persona-, demographic-, survey-, and interview-based, and composite); we analyse four — composite, interview-only, survey-only, demographic — and leave the persona-based condition outside our re-identification measurement (data/PROVENANCE.md; FIGURE2_PIPELINE.md). Human retest ceiling 96.8 %.

No new data was collected. The attack uses only already-published content.

4.2 Methods compared

Twelve heterogeneous methods carry the quality-leakage analysis: 8 LLM twins and 4 statistical reference points (Demographics Only, B1 argmax over 14 demographics, B2 argmax, PMM $k=10$). Additional comparators appear in § 5.2: logistic regression on context, a $k=1$ donor on context, two individualised *generators* (sampled LR, Gaussian copula), and an explicitly advantaged $k=1$ donor fitted directly on the target block. Sequential CART and a 300-tree forest were tried before that donor and abandoned at 0.45–0.53 accuracy: the retained comparator was not chosen to lose [c7-synth-ajuste-resultats.md § 1].

4.3 Metrics

Attacks. Two matching rules are used throughout, and every rate is labelled with the one that produced it. The **naive** attack ranks candidates by Hamming-type agreement over common items. The **strong** attack (A-LLR) ranks them by a rarity-weighted log-likelihood whose parameters are estimated **out-of-fold over 5 folds**, so no candidate is scored with parameters fitted on itself [c7-attaquant-fort-resultats.md]. A third variant (A-MI), weighted by measured per-item information, scores higher still on the Park archive (93.25 %) but estimates its weights **in sample** — an advantage declared in the preregistration — so we never report it as our result. An **adaptive** attacker, used only against the defense (§ 6.1), additionally knows the defense mechanism and which items it leaves untouched.

Closed-world top-1 and top-10: rank of the true target among all N candidates by the rule stated. *Open-world:* true-positive rate against false-positive rate, where the system may decline to accuse (§ 5.3). *Bits of identity:* $\log_2$ of the ratio of correct one-shot guessing probabilities before and after conditioning on the persona — a Rényi min-entropy leakage quantity [34]. *Imitation quality:* fidelite_plancher, the share of the human test-retest floor attained.

All intervals are 95 % bootstrap confidence intervals unless stated.

4.4 Preregistration and self-refutation

Attack, defense, mechanism, Park et al. archive replication and the disjoint-item test were each preregistered before computation. We report every refuted prediction in Table 1. A census of the repository’s preregistrations found **no inversion** between a plan and its result across the 28 verifiable pairs; for the large majority the ordering is provable from version-control history, while a small number were committed alongside their results or left unversioned, which we state as a limit rather than claim an unbroken chain. The plans are not yet third-party timestamped.

4.5 Validity controls

Provenance. On Twin-2K-500, the context contains 0 wave-4 columns and 0 wave-4 QIDs; maximum normalised mutual information between context and item is 0.16 against 0.31 for the retest on the same item. On cells where wave 4 differs from waves 1–3, the JSON 4.1 output equals wave 4 in 37.7 % of cases and waves 1–3 in 37.9 % — symmetric, so no indication of copying. The randomisation channel is excluded [c7-contre-examen-2026-09-11.md § 2].

Provenance, Park archive. Over 34,915 cells where the human changed their mind, the composite follows wave 1 in 41.6 % and wave 2 in 43.2 % — a difference of –1.6 points, no bias. The 4.3–4.9 point wave-1 bias of the other three conditions holds for the demographic condition too, so it is a shared calibration artifact, not item leakage [c7-stanford-provenance-resultats.md].

Empty-run validation. Uniform ranks yield 0.00 bits; the B0 mode baseline yields –0.004 [–0.003 ; 0.002] bits.

5 Results

5.1 The coupling, and the null that absorbs it

Across 12 heterogeneous methods, the ordering by imitation quality and the ordering by re-identification rate nearly coincide: Spearman 0.958 [0.930 ; 0.993], $n = 12$, on the same 12 items [c7-compromis-resultats.md § 1]. It is robust: restricted to LLMs over a narrow fidelity range, $\rho = 0.976$ [0.952 ; 1.0]; jackknife 0.958–0.986 (§ 4 of the same source).

The preregistered disjoint-item test measures fidelity on one half of the 60 items and leakage on the other, over 50 stratified random A/B splits: Spearman **0.969**, median 0.972, 95 % CI [0.937 ; 0.993], with 100 % of splits above 0.7 [c7-disjoint-resultats.md § 1]. Prediction (a) confirmed — the relation is not an artifact of recycled computation.

Prediction (b) was refuted, and the witness that refutes it has since been rebuilt to remove a real defect without changing the verdict. The original marginal null did carry that defect: its *wrong* cells avoided the person’s true answer (c7_disjoint.py l. 133), which inflated its leakage by a factor 1.3 (top-1 41.2 % against 31.6 % once wrong values are drawn from the population marginal). With the defect corrected, the witness still reproduces the coupling — **$\rho = 0.974$, 5th–95th percentiles [0.950 ; 0.993], against 0.965 observed** over the 12 configurations, measured on whole items (the human retest point excluded; the 0.969 above is the disjoint-split value of the same quantity; Figure 2) — so **prediction (b) remains refuted**: reproducing the correlation between imitation quality and re-identifiability requires nothing more than a matched per-person accuracy and correct positions drawn at random.

This witness is not an information-free object, and we do not describe it as one: at matched accuracy it leaks **more** than our twin (31.6 % top-1 against 20.73 %). Five further witnesses, which match that accuracy against a *surrogate* target rather than against the person, fall to chance on both axes (top-1 0.040–0.063 % against 0.049 %, 0.00 bits, accuracy against the truth 0.41–0.44 against 0.53); the observed ρ exceeds them, but that margin bears on nothing, their null distribution being that of Spearman under total absence of information at $n = 12$ (95th percentiles 0.547–0.594 against 0.497 expected).

We therefore do not claim, anywhere in this paper, that a twin’s fidelity to its person causes its leakage, nor that fidelity and identifiability are one axis. We also do not claim that at equal accuracy every twin leaks more than every predictor: the residuals of the regression on accuracy do **not** separate the families, with 3 of 7 LLM configurations falling below the line.

One positive point remains testable outside this circularity: the human point, in raw position, dominates on both axes — fidelity 0.303, leakage 0.520 — against 0.154 and 0.067 for the best twin. There is no saturation at the top of the range.

Limits. Twelve points. Raw accuracy is only a noisy proxy ($r = 0.72$, $p = 0.008$) for the right axis. The marginal null has only 12 configurations, which makes Spearman coarse — the null touches 1.000 through small-sample effects — so the verdict concerns the order of magnitude, not the third decimal.

5.2 Closed-world rates and the comparator question

At comparable marginal accuracy, no non-LLM predictor we tested exceeds 0.3 % top-1 where the twin reaches 20.7 % [19.1 ; 22.5] under the naive attack, and **23.23 % [21.5 ; 25.0]** under the strong one [c7-attaquant-fort-resultats.md] — **in a closed world, with the target always present in a pool of 2,058**. The comparator figures below are naive-attack rates, which makes the contrast conservative. JSON Persona GPT4.1 attains that rate at accuracy 0.590. The comparators, at accuracy 0.457–0.511: PMM $k=10$, 0.23 % [0.06 ; 0.42]; B2 argmax, 0.07 % [0 ; 0.19]; $k=1$ donor on context alone, 0.13 %; LR on context, 0.22 % [c7-contre-examen-2026-09-11.md § 1]. Two individualised *generators* — sampled LR and Gaussian copula — give 0.25 % [0.11 ; 0.41] and 0.22 % [0.11 ; 0.36] [c7-generateur-resultats.md], answering the objection that we compare an individualised generator against predictors that regress to the mean.

We then gave a comparator an explicit advantage: a $k=1$ nearest-neighbour donor fitted **directly on the target block’s answers** — seeing what the LLM twin never sees. It reaches accuracy 0.584 against the twin’s 0.590, carries genuine individual signal (accuracy drops 0.142 under within-segment permutation, 0.584 true against 0.442 permuted), and despite that advantage obtains a top-1 of **0.00 % [0 ; 0.18]** (Clopper-Pearson, $n = 2,058$) [c7-synth-ajuste-resultats.md § 1].

This claim does not generalise to every leakage metric, and top-10 runs the other way. That same advantaged comparator reaches a top-10 of **53.2 % [51.0 ; 55.3]**, above the twin’s 42.7 % [c7-synth-ajuste-resultats.md § 2]. The direction of the whole contrast therefore depends on the choice of k , which no principle fixes at 1.

We do not claim that all individualised prediction identifies, nor strict accuracy equality: the generators plateau at 0.476, never 0.590. Neither synthpop/CART nor CTGAN was tested.

5.3 Open world — the defensible measurement

Removing the closed-world assumption gives the number we consider hardest to attack in review.

At a false-accusation rate of 1 %, the strong attack (A-LLR, parameters estimated out-of-fold over 5 folds) recovers the right person **60.17 % [54.5 ; 64.4]** of the time on the Park archive and **4.28 %**

[3.3 ; 5.6] on Twin-2K-500, under a person-level bootstrap that re-estimates a_j and q_j in each resample while keeping every copy of a person in the same fold [c7-fort-monde-ouvert-ic-2026-09-12.md § 2]. At FPR = 0.1 % these fall to 44.37 % [23.3 ; 53.8] (Park) and 1.01 % [0.05 ; 2.82] (Twin) — intervals this wide because the threshold there rests on roughly one absolute false positive on Park and two on Twin, so we report them as very unstable estimates, not as measured rates. Statistical comparators (Demographics Only, PMM) stay indistinguishable from noise at both FPRs. The human ceiling at FPR = 1 % is 54.5 % [50.8 ; 57.5] and 90.7 % [88.2 ; 93.3] [c7-monde-ouvert-ic-2026-09-12.md § 1].

These figures replace the ones we first published, and the correction runs in both directions. The naive attack gave 3.04 % [2.0 ; 4.0] and 20.39 % [15.7 ; 24.5] at 1 % FPR, and 0.93 % [0.2 ; 1.6] and 8.47 % [2.5 ; 16.2] at 0.1 %. On the Park archive the strengthening is real and statistically clear — no overlap at either FPR, and the rate triples (20.39 → 60.17 %). On Twin-2K-500 the apparent gain does not survive its interval: 3.04 → 4.28 % and 0.93 → 1.01 % both overlap broadly, so we do not present the strong attacker as identifying better than the naive one here. Park's closed-world gain (65.51 → 90.40 %) is a different quantity, untouched by this comparison.

Our preregistered prediction was refuted, and one half of it was then reversed by a better attack. We announced > 5 % (Twin) and > 30 % (Park). Under the naive attack both fell below the bar — the refutation we reported. Under the strong attack Twin (4.28 %) still falls below, while Park (60.17 %) passes the bar it had failed. We state both, in that order.

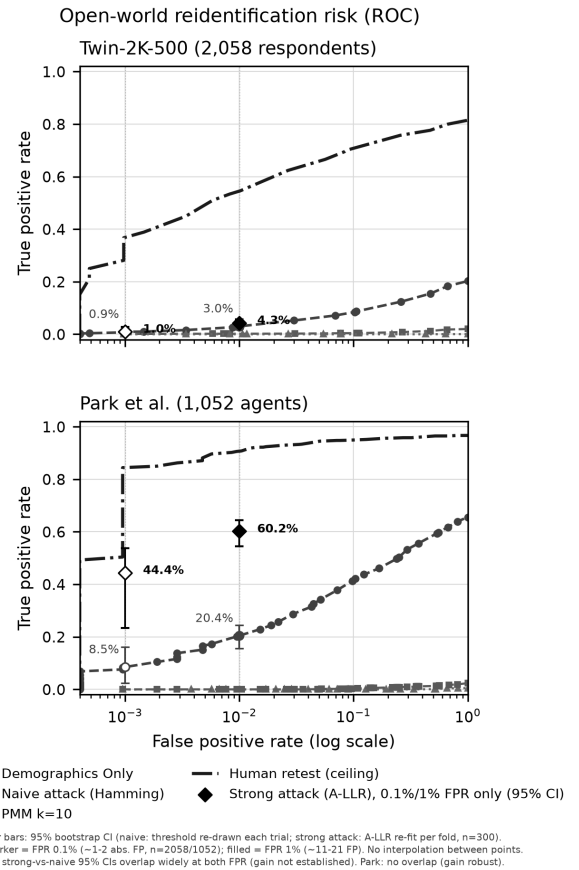


Figure 1: Figure 1 — Open-world detection, naive and strong attack. Two panels, one per dataset, false-accusation rate on a log scale against true-detection rate. Each overlays the naive Hamming attack as a full ROC curve with the strong A-LLR attack as isolated diamonds, the latter measured at only two thresholds (FPR = 0.1 % and 1 %): no curve should be read into the diamonds. Statistical comparators sit near zero, the human retest ceiling above. At a glance: at a tenable false-accusation rate the risk is real and far above the comparators under both attacks, and the strong attack lifts the Park archive close to the human ceiling. This figure depends on no causal reading of the quality-leakage coupling. Data: resultats/c7-monde-ouvert-roc.csv, resultats/c7-attaquant-fort.csv.

5.4 The twin-to-twin channel within one pipeline — and T2 itself, refuted

Two twins of the same person designate each other when they share enough items, with no real data on the attacker's side. The measurements immediately below share one team and, for Twin-2K-500, one persona source per person: they gauge the channel within a pipeline, not the cross-organisation threat model T2 (§ 3), tested at the end of this section and refuted.

Twin-2K-500, tabulated by common-item count [c7-transfert-resultats.md]. 30 pairs sharing 60 common items: mean top-1 **36.4 %**, maximum 83.58 %. 12 pairs sharing 19 common items: mean top-1 **0.45 %**, minimum 0.06 % (chance 0.05–0.10 %). Demographics Only against the rich twins gives 7.76 %, clearly below 36.4 %. The anti-artifact control — a decoy from the same segment — gives 0.04 % on the 60-item pairs (factor 900) and 0.10 % on the 19-item pairs (factor 4.5).

Park archive replication [c7-transfert-standford-resultats.md]: 1,052 agents, 177 common items. Interview→survey top-1 **11.9 %** [10.1 ; 13.8]; survey→interview **13.0 %** [11.2 ; 14.9]; top-10 46.4 % [43.6 ; 49.4] and 45.2 % [42.3 ; 48.1] respectively [c7-transfert-standford-attaque.csv]. Demographic bound 1.5–2.5 %. Anti-artifact decoy control **at most 0.31 %**.

We never report a single figure for this channel without its common-item count, within or across datasets. A previously circulated average of 26.11 % pooled the two item regimes and is withdrawn. Nor do we claim leakage crosses waves: that arm is infeasible, with 0 items in common between the 108 wave-4 items and the 494 context items.

Preregistered prediction refuted. We set a threshold of at least 20 % top-1 on the Park archive; the observed 11.9–13.0 % does not reach it. The preregistered *failure* criterion (“does not clearly exceed the demographic bound”) is however not met either, at a factor of 5–8×: the channel replicates, more weakly than on Twin-2K-500, below the numeric threshold we predicted.

The channel is not an artifact of a shared prompt template or of demographics. Decomposing agreement between two twins of the same person over the 60 common items: the floor between *strangers* is **46.0 %**, the demographic-ideological segment adds only **3.8 points**, and the person adds **17.4 points** — 82 % of the rise above the floor. A pure-population witness (each twin replaced by its segment’s leave-one-out modal answer) identifies the right person **0.07 %** of the time against **36.4 %** for the real attack [c7-temoin-prompt-resultats.md].

The cross-dataset anomaly, explained in part. The Park archive shares 177 common items — roughly three times the 60 of Twin-2K-500 — and leaks roughly three times *less*. Item count is the wrong denominator: the 177 GSS items are strongly redundant (mean Cramér’s V 0.1055 against 0.0834), giving $\approx$ **9.0 effectively independent items against $\approx$ 10.1** on Twin — a real ratio of **0.89**, **not 2.95** [c7-anomalie-park-2026-09-12.md § 4]. Pool size runs the wrong way too: shrinking Twin’s pool to 1,052 *raises* its top-1 (37.3 → 44.1 %, disjoint intervals), while Park already has the smaller pool. **A residue of roughly 3× remains unexplained** at comparable effective information, and we declare it: the remaining candidates — persona format and length, and the different generating pipelines — are not testable on the material we hold.

Limits. The 12 low pairs are exactly those at 19 common items: the effect follows the shared-item count, not an aberrant configuration. On the Park archive the item-rate relation does not follow Twin’s.

T2 itself: two genuinely independent pipelines, tested and refuted. Two pipelines sharing nothing but the target person — deepseek-v4-f1ash structured JSON dossiers versus qwen-2.5-72b narrative biographies, different templates, different output formats — matched head to head on the same 60 items ($n = 142$ of 200 planned, stopped by a second OpenRouter transport error; a first

such stop, at $n = 82$, was reconciled and confirmed unbilled) [c7-deux-organisations-resultats.md]. Preregistered rule: T2 survives if the top-1 CI excludes the segment control *and* stays at least 2× the demographic baseline. Result: top-1 **1.8 %** [0.4 ; 3.6], below the baseline alone (9.2 %, let alone 2×) and the segment decoy (25.5 % [19.9 ; 31.3]) — lower still than at $n = 82$ (3.7 %). **T2 is refuted**; the 36.4 % figure above is a same-pipeline measurement, not evidence for it.

5.5 Mechanism: being right while deviating — and three refuted hypotheses

What identifies is neither the frequency of departures from the modal answer nor answer diversity, but their **correctness** — and the whole response vector is required.

Correctness of deviations: 68.5 % for the human retest (leakage 81.6 %), 50.3 % for JSON 4.1 (20.6 %), 43.3 % for Demographics Only (2.14 %), 39.1 % for PMM (0.22 %). PMM deviates (49.7 %) and diversifies (0.560) as much as the twin without leaking [c7-deviations-resultats.md].

The ablation is the strongest evidence: permuting the order of each twin’s 40 purchase answers drops closed-world top-1 from **33.1 %** [31.2 ; 35.2] to **0.046 %** [0 ; 0.11], below chance. An oracle given only the count of “yes” answers gives 0.29 % [c7-mecanisme-resultats.md H3]. The identifying quantity is the dependence structure between answers, carried by the conditioning — not any answer taken in isolation.

That ablation starts from a higher rate than this paper’s headline 20.7 %, and the difference is a change of attack surface, not a discrepancy: the ablation attacks the **40 purchase items alone**, where the headline rate uses all **60 common items**. Adding the 20 opinion items therefore *lowers* top-1, from 33.1 % to 20.7 %. That is consistent with the rest of this section rather than in tension with it — those items identify 46× less, and a naive Hamming rule is diluted by uninformative items rather than helped by them — and we read it as an independent confirmation of H1’s direction, not as a loose denominator.

Three preregistered hypotheses refuted. H1 (entropy): opinion items carry 2.10 bits against 0.99 for purchase items and identify 46× less — top-1 0.24 % [0.07 ; 0.46] for the 20 opinion items against 11.0 % [9.9 ; 12.3] for the purchase subset matched to them on entropy by Hungarian assignment — a 20-item comparison, not the 40-item block above. H4 (stereotypy): 36.2 % [35.6 ; 36.8] against 37.5 % [36.9 ; 38.1] in humans. And “deviations alone carry $\geq$ 80 % of the leakage”: 0.68 % in the mixed condition.

We do not present “wrong deviations alone = 0.0 %” as a result: it is a floor of the method, which cannot detect a signature there.

Limits. The mechanism is established on Twin’s purchase block alone, and is in apparent tension with the Park archive, where opinion items identify most. Two simple explanations are excluded: at 20 items the Park archive gives 11.7 % against 0.24 % for Twin’s 20 opinion items, **at lower per-item entropy** (1.30 against 2.10), so neither item count nor entropy accounts for it [c7-standford-resultats.md § 6]. What remains is consistent with the block effect, which we present as the **best-supported hypothesis, not a result**: the dedicated test (copula alone against margins alone) is not done.

5.6 Bits of identity: the instrument transports, the rate does not

An LLM twin costs approximately 0.4 bits of identity per point of accuracy gained, against 0.068 for the best useful statistical comparator. Twin JSON 4.1: **3.55 bits** [3.39 ; 3.72] against a ceiling of $\log_2 N = 11.01$. Park composite: **7.47** [7.27 ; 7.66] against a ceiling of 10.04. PMM: 0.19 [0.16 ; 0.23]. B2: 0.08 [c7-bits-resultats.md § 1–2].

Transportability is the point. Bits normalised by human entropy give 0.0439 (Twin) against 0.0327 (Park) — a factor of **1.34** — where top-1 varies by a factor of **3.2**. Demographics Only and PMM leak with **no accuracy gain at all** (−0.5 and −2.4 points), a case that a ratio alone would mask.

A nuance against our own hypothesis. The human retest has a ratio of 0.37 (Twin) and 0.44 (Park) — as good as or better than the twins. The excess cost exists relative to statistical predictors, **not relative to a human answering twice**. We therefore do not claim that LLM twins leak more than anything else.

Limits. The bits are a **lower bound** (dyadic bound plus Miller-Madow). The sum of per-item mutual information is unusable as a total: 58.7 bits against a ceiling of 10.04 on the Park archive, as preregistered.

5.7 The risk depends on the recipe — and we could not reproduce it

Our own twins, produced in a single call to cheap models from a raw profile, identify almost nobody: 0.00 % to **0.83 %** [0 ; 2.15] (deepseek-v4/R1) across 7 cells, against 2.13 % for the other team’s Demographics Only. The interval [0 ; 2.15] **contains** 2.13 %, so no precise factor is established — only the order of magnitude, that no model reaches 1 %, is defensible [c7-gen-resultats.md]. Any previously circulated “25× below” figure is withdrawn.

Memorization is not the explanation here: llama31-8b, with a training cutoff preceding publication, copies nothing verbatim and leaks no more than the models that do copy. This is a control on our own twins, in a regime where none of them identifies almost anyone; we do not extend it to the published twins of § 5.2–§ 5.4, whose training data we cannot probe (§ 1.2).

Two further preregistered predictions refuted. (i) $R1 \geq 10$ % on at least 2 of 3 models — no model reaches 1 %. (ii) Call granularity would explain the leakage — not confirmed (underpowered, especially at $n = 10$ for the per-item arm): single call 0.00 % [0 ; 8.8] and per-item call 0.00 % [0 ; 30.85], with top-10 running in the **opposite** direction to the prediction (7.5 % against 0 %) [c7-recette-resultats.md].

We therefore do not claim that leakage comes from the call format, nor that the cause is identified. **It is not.**

We then paid to test the obvious explanation, and it failed. Raising fidelity with a frontier model to see whether the leakage returns was run on openai/gpt-4.1 (Twin’s own model class) using Twin’s per-item call recipe without modification: 30 persons × 60 items = 1,800 calls, 100 % parse rate, 4.9949 USD of actual cost [c7-fort-resultats.md]. Accuracy reached **0.4722** [0.4361 ; 0.5050] against Twin’s 0.574; fidelity **0.1714** against their 0.708; top-1 and top-10 **0.00 %** [0 ; 11.57] (Clopper-Pearson, $n = 30$) — below even

our cheap twins (0.83 % at best). **One further preregistered prediction refuted:** accuracy > 0.55; top-1 > 5 % is **not refuted** — 0/30 gives an exact 95 % CI of [0 ; 11.57 %], which contains the 5 % threshold: inconclusive for lack of power. Fidelity > 0.10 was confirmed at 0.1714, but barely above our best cheap twin (0.177): the frontier model bought us almost nothing. We did not make the leakage come back, because we did not manage to reach Twin’s fidelity — and it is that fidelity gap, not the underpowered top-1, that carries the conclusion.

The curve is confirmed a third time, by an independent point. At the fidelity actually reached (0.1714), the regression over the 12 points of c7-compromis.csv (slope 0.1834, $r = 0.785$) predicts leakage of **1.10 %**, 95 % CI [−9.15 ; 11.35], and the observed 0.00 % falls inside it. Coming from a new model *and* a new recipe, it is an independent check: fidelity never rose, so the point follows the curve rather than breaking it.

The unknown narrows instead of disappearing. Neither the model nor the call granularity, what remains implicated is the **format and length of the profile**: Twin’s full JSON persona runs to roughly 121,000 characters against our 8,000-character truncation. Only Twin’s exact pipeline or their complete personas would settle it.

Limits. $n = 40$ persons for the recipe arm, truncated to $n = 10$ on the per-item arm, and $n = 30$ on the paid frontier-model arm; a floor effect is not excluded. Roughly 1,490 calls first failed with HTTP 402 from in-flight credit reservation, not lack of balance, and were retried at decreasing concurrency without double billing.

5.8 Replication on a second dataset

On the Park archive, the composite agent designates the right participant among 1,052 **in a closed world** in **90.40 %** [88.6 ; 92.2] of cases under the strong attack, with a top-10 of 97.8 % [c7-attaquant-fort-resultats.md]. The human retest ceiling is 96.8 %.

This figure replaces the 65.51 % [62.7 ; 68.3] **we first published** (measured at 65.7 % [62.7 ; 68.6] by the original script — same condition, two implementations): a naive agreement attack understated this archive’s leakage by 38 % in relative terms. An in-sample information-weighted variant reaches 93.25 %, which we do not claim as our result (§ 4.3).

The remaining conditions are **naive-attack measurements**, and the strong attack was **not** re-run on them: interview-only **44.7 %** [41.8 ; 47.5], survey-only 20.6 % [18.2 ; 23.0], demographic 2.26 % [c7-stanford-resultats.md]. The interview-only figure carries zero documentary exposure to the survey, so it is the cleanest demonstration that an agent built from a conversation alone identifies its person — but given what the strong attack did to the composite, **44.7 % must be read as a lower bound.**

We do not compare the Park and Twin rates as measurements of the same thing: item count (177 against 60), per-item entropy (1.30 against 0.99) and human ceilings (96.8 against 81.6) all differ. Equalised at $k = 60$, Park gives 34.0 % [23.5 ; 51.5] against Twin’s 20.7 % — naive attack, 20 draws, a very wide interval.

5.9 Scale

From $N = 50$ to $N = 2,058$, the twin/demographic ratio grows from 2.8 to 9.7: top-1 falls from 53.2 % [50.0 ; 56.3] at $N = 50$ to 20.7 %

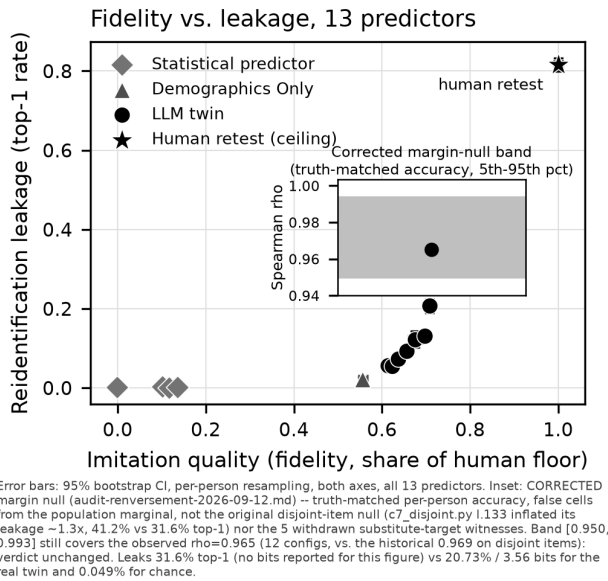


Figure 2: Figure 2 — Imitation quality against leakage, with its control (article/figures/fig2-couplage.png). Abscissa: individual fidelity (fidelite_plancher, share of the human floor, 0 to 1, linear). Ordinate: closed-world top-1 leakage, linear, 0 to 0.85. Thirteen points: the LLM twins (filled circles), the statistical reference points (grey diamonds), Demographics Only broken out (triangle), and the human retest (star) at fidelity ≈ 1 / leakage 81.6 %. All thirteen points carry error bars on both axes, from a per-person bootstrap with 2,000 resamples. Inset: the marginal null on a Spearman-rho axis — the 5th–95th percentile envelope of the corrected witness (mean rho 0.974) matched only on per-person accuracy (c7-disjoint-nul.csv) drawn as a grey band, with the observed rho (0.965) plotted as a single point inside that band. *What the reader should see at a glance:* the points rise together from left to right with no plateau at the top, but the observed correlation sits inside the band that a null matched only on accuracy already produces. An honest contribution: the coupling is measured, not shown to be individual-specific. Data: resultats/c7-compromis.csv, resultats/c7-compromis-robustesse-points.csv, resultats/c7-disjoint-nul.csv.

[20.7 ; 20.8] at $N = 2,058$, and the share of the human ceiling from 57.0 % to 25.4 % [c7-echelle-resultats.md]. These are naive-attack rates; the scale study was not re-run under the strong attack.

We extrapolate no value beyond $N \approx 4,000$: a power law and a logarithmic law fitted on the same 6 points already diverge at twice the range (18.1 % against 13.9 %).

6 Defense

Shuffling the purchase answers between people of the same demographic segment (D4) brings closed-world top-1 from 20.68 % [19.04 ; 22.41] to 0.13 % [0.01 ; 0.28] [c7-defense-resultats.md].

Per-item distribution and between-segment differences are preserved **exactly, by construction**, as PRAM predicts.

The cost is not a single average. Reported by component: 0.0 points on the per-item distribution, 0.0 points on group differences, and **4.4 points on inter-item correlations**. We therefore withdraw the summary that presented D4's cost as a single mean of 1.47 points: that figure is the unweighted mean of the three components (utilite_globale in c7-defense-courbe.csv), and quoting it alone divides by three an effect that falls entirely on one of them. The quantity itself is not withdrawn, only its name and its use: it is a **composite utility index**, and it is the one form in which the comparison with differential privacy can be made at all, since the DP mechanism's loss is measured on that same index (§ 6.2). Wherever it appears below it is named as the index, never as the cost of the defense.

The cost is worse than that summary suggests. On that same indicator measured against the real human answers, erreur_correlations_hum rises from 5.78 to 9.71 after D4, a degradation of 68 %. The defended twin moves away from the humans on correlations; it does not move closer. Any framing in which the defense's cost is absorbed by an error already present compares an increment to a level, and is incorrect.

Alternatives: D1 ($k=10$ aggregation) gives 0.55 % for 3.8 points spread across all three components; D2 (noise) never descends below 1 %.

6.1 The defense against a strong, then an adaptive attacker

The defense figure above was measured against the naive attack, which § 5.8 shows can badly understate leakage. We therefore re-ran it against the strong attack, recalibrated on the *defended* outputs, and then against an attacker who knows the defense mechanism and knows which items it leaves untouched [c7-attaquant-fort-resultats.md].

D4 holds. Top-1 goes from 0.13 % (naive) to 0.24 % under the recalibrated strong attack, and the **adaptive** attacker plateaus at 0.29 % (strategy S1, leaving the 20 opinion items intact; S1+S3 identical; segment-invariant strategy S3 alone 0.05 %). That is two orders of magnitude below the 20.69 % undefended rate, and never above 1 %. **Attribute disclosure is not demonstrated either:** the best strategy names the correct S_{gra} segment in 7.7 % of cases, against 6.8 % at chance and 12.6 % for the trivial rule of always answering the most frequent segment (40 segments) — the attack does worse than not attacking.

A reservation that bounds the guarantee. This residue comes *entirely* from the 20 opinion items D4 does not permute. The guarantee holds for this split — 40 purchase items shuffled, 20 opinion intact — not for a design leaving a more informative block untouched. We do not present 0.29 % as a bound on the defense in general.

6.2 Differential privacy: our prediction refuted, and the argument that survives

We preregistered the prediction that at a moderate budget, differential privacy would be dominated by D4 on the aggregate risk-utility table. **It is not.** With per-item marginals perturbed by Laplace noise

(L1 sensitivity 2, budget split equally over 60 items, basic sequential composition) followed by i.i.d. per-item sampling — degree-0 PrivBayes, no joint structure — the DP synthesiser at $\epsilon = 3$ and $\epsilon = 10$ loses **3.33** and **3.38** points on the **composite utility index**, within 5 points of D4’s **1.47** on that same index, while its top-1 (**0.158 %**, **0.000 %**) is **not above** ours (0.126 %) [c7-dp-resultats.md]. The comparison runs on the aggregate index because that is the only scale on which the two mechanisms are commensurable; 1.47 is not D4’s cost, which falls on a single component (§ 6). We do not claim our defense beats DP, on either axis.

The utility floor is architectural, not budgetary: the non-private control ($\epsilon = \infty$) still loses 3.52 points, only distribution error falling with epsilon ($9.7 \rightarrow 3.2$ points).

What survives is a theoretical point, not an empirical win. DP protects an individual’s *membership* in the dataset used to compute a published statistic — “is Alice in the sample?”. Our attack assumes Alice is already known, since her profile is the twin’s input, and asks whether **the output conditioned on Alice** can be linked back to her: record linkage, not membership. The DP generator escapes that attack only by **never conditioning on an individual** — individual fidelity stays at most 0.2 points at every budget, including infinite — that is, by refusing the twin’s task altogether. Its low leakage is a by-product of that incapacity, not of the budget chosen. A panel holder who needs population statistics should use DP; one who needs a per-person twin cannot obtain one from this mechanism at any epsilon.

Limits. Written by hand rather than with a reference library, and not full PrivBayes; epsilon covers only the published histograms, with no composition across the other analyses in this repository; correlations, between-segment differences and the S_{gra} covariate are not protected.

6.3 What the defense costs a downstream analyst

A defense is only usable if the analyses people actually run survive it. We measured three on the same 2,058 persons [c7-utilite-aval-resultats.md].

Preserved exactly. Group comparisons are **identical to 16 decimal places** between the raw and defended twin — shuffling within S_{gra} moves no group mean — and demographic regression coefficients (gender, age) keep their sign and significance.

Destroyed. Purchase-item coefficients that were significant in the raw twin ($p < 0.01$ and $p < 0.001$) become non-significant after D4. On a PCA of the 40 purchase items, **45 % of the first component’s loadings are inverted** relative to humans, against **0 %** for the raw twin.

The honest comparison. The unprotected twin was already wrong: it inverts the sign of the male-female gap (+0.010 against −0.047 in humans) and already loses 5.3 points of PC1+PC2 variance (15.0 % $\rightarrow$ 9.8 %). D4 adds **3.5 points** (9.8 % $\rightarrow$ 6.3 %), less than the error already present — but the axis inversion is **D4’s own doing**. It changes *which* items compose the structure more than it changes the structure’s strength.

Practical recommendation. With a D4-defended twin, group analyses and predominantly demographic regressions remain reliable. What becomes unusable is anything resting on the link between two answers of the same person — inter-item regression, the composition of a PCA axis.

Limits. Tested on a single block, of a single twin, of a single dataset (§ 2.6 places the mechanism in the PRAM and data-swapping tradition).

7 Discussion and Limitations

7.1 The fifteen refuted preregistered predictions, and two inconclusive

Rows 9 and 16 were listed as refutations until adversarial review found that a percentile bootstrap on a zero-event sample can only return “[0 ; 0]” — an arithmetic property of the resampling, not a confidence interval. Recomputed exactly, neither is refuted: both are inconclusive for lack of power, and we report the reclassification rather than keep two refutations we are not entitled to.

Three further preregistered outcomes were partial rather than refuted, and we count them as neither confirmations nor refutations: D4 erases two publishable inter-item effects while leaving signs unchanged, and its PCA prediction holds on variance but is contradicted on the loadings (§ 6.3); the paid frontier-model run cleared its fidelity > 0.10 bar at 0.1714, barely above our cheapest twin (§ 5.7). One prediction held outright under adversarial pressure: P2, that D4 stays below 1 % against a strong attacker (§ 6.1).

7.2 What we do not know

An excess cost in bits per unit of fidelity, real but unexplained. On equal samples, $CV(\text{bits/fidelity}) = 0.436$ against $CV(\text{bits/accuracy point}) = 0.357$ over the same 9 points: the second remains less dispersed, but the gap shrinks. With so few points and no identified mechanism, the excess **must not be attributed to “LLMs” as a family**; we describe it as unexplained [c7-compromis-resultats.md § 5].

We did not reproduce the leakage with our own twins (§ 5.7), including with a paid frontier model on Twin’s own per-item recipe — an open limit, never a mechanism, and a narrower one than before: neither the model nor the call granularity, leaving the persona’s format and length. Only Twin’s exact pipeline or their complete personas would settle it.

Rates do not predict across datasets. The normal-maxima model lands close on Park $k = 60$ (0.354 predicted, 0.340 observed) but fails both its controls (0.053 for 0.206 observed; 0.983 for 0.656): an isolated success between two failures is a coincidence, not a law [c7-bits-resultats.md § 3]. Only conditional information retains sign and order of magnitude across datasets ($r = 0.72$ and 0.89).

The twin-to-twin channel’s cross-dataset rate is only partly explained (§ 5.4): item count is the wrong denominator, pool size runs the wrong way, and a residue of roughly $3\times$ survives both controls with no mechanism we can test on the material we hold.

7.3 Multiplicity

Most of this paper’s central results are bootstrap confidence intervals around a descriptive measure — re-identification rate, rank

Table 1: The seventeen preregistered predictions and their outcomes (§7.1).

#	Prediction (preregistered)	Outcome	Source
1	The quality-leakage coupling exceeds a null matched on accuracy alone	Refuted. Null rho 0.974 [0.950 ; 0.993] against 0.965 observed, over the 12 configurations on whole items; the original null (0.984 against 0.969, disjoint splits) carried a defect since corrected, with no change of verdict (§ 5.1)	audit-renversement-2026-09-12.md
2	Open-world > 5 % (Twin) and > 30 % (Park) at FPR = 1 %	Refuted as first measured (naive attack: 3.04 % and 20.39 %). Under the strong attack Twin still fails at 4.28 %, while Park passes at 60.17 %	c7-monde-ouvert-resultats.md, c7-attaquant-fort-resultats.md
3	Twin-to-twin top-1 ≥ 20 % on the Park archive	Refuted. 11.9–13.0 % at 177 common items (failure criterion not met either)	c7-transfert-stanford-resultats.md
4	The twin-to-twin channel leaks on the 19-common-item pairs	Refuted. 0.45 % mean top-1, at chance (0.05–0.10 %)	c7-transfert-resultats.md
5	H1: per-item entropy drives identification	Refuted. Opinion items 2.10 bits, identify $46\times$ less	c7-mecanisme-resultats.md
6	H4: twins are more stereotyped than humans	Refuted. 36.2 % [35.6 ; 36.8] vs 37.5 % [36.9 ; 38.1]	c7-mecanisme-resultats.md
7	Deviations alone carry ≥ 80 % of the leakage	Refuted. 0.68 % in the mixed condition	c7-deviations-resultats.md
8	R1 ≥ 10 % on ≥ 2 of 3 of our own models	Refuted. No model reaches 1 %	c7-gen-resultats.md
9	Call granularity explains the leakage	Not confirmed — inconclusive (underpowered at n = 10). 0.00 % both arms (95 % CI [0 ; 8.8] at n=40, [0 ; 30.85] at n=10); top-10 runs opposite (7.5 % vs 0 %)	c7-recette-resultats.md
10	Cost per unit of individual fidelity is roughly constant	Refuted at equal sample. CV 0.436 vs 0.357 on the same 9 points	c7-compromis-resultats.md § 5
11	Per-item entropy correlates with identifying power consistently	Refuted. Opposite sign by dataset: $r = -0.81$ (Twin), $+0.57$ (Park)	c7-bits-resultats.md § 3
12	P1: a stronger attacker gains ≥ 20 % relative over the naive attack	Refuted on Twin (+12.2 %, 20.69 $\rightarrow$ 23.23 %); held on Park (+38.0 %, 65.51 $\rightarrow$ 90.40 %)	c7-attaquant-fort-resultats.md
13	P3: an adaptive attacker knowing the mechanism breaks the defense	Refuted, in the defense’s favour. Plateaus at 0.29 %, never above 1 %	c7-attaquant-fort-resultats.md
14	At a moderate budget, DP is dominated by our defense on the aggregate table	Refuted. eps = 3/10 lose 3.33/3.38 points on the composite utility index against D4’s 1.47 on that same index (not D4’s per-component cost, § 6), top-1 0.158 % [0.012 ; 0.340] / 0.000 % [0 ; 0.18] against 0.126 % [0.012 ; 0.284]	c7-dp-resultats.md
15	A frontier model on Twin’s per-item recipe reaches accuracy > 0.55	Refuted. 0.4722 [0.4361 ; 0.5050]	c7-fort-resultats.md
16	That same twin reaches top-1 > 5 %	Not refuted — inconclusive. 0.00 % [0 ; 11.57] (Clopper-Pearson, n = 30); the preregistered 5 % threshold lies inside the interval	c7-fort-resultats.md
17	T2, independent pipelines: top-1 CI excludes the segment control and stays $\geq 2\times$ the demographic baseline	Refuted. 1.8 % [0.4 ; 3.6] at n = 142/200 (API transport error); below the baseline alone (9.2 %), let alone $2\times$, and below the segment control (25.5 %)	c7-deux-organisations-resultats.md

correlation, bits of identity — not classical hypothesis tests; of the 34 preregistered predictions underpinning those claims at the time of the census, 11 were refuted and 3 judged inconclusive, a rate of roughly one third, the opposite of the signature of data dredging. Of the 3 tests carrying a classical p-value, a Holm correction leaves the two smallest standing (0.002 and 0.008, adjusted to 0.006 and 0.016) and confirms the non-significance of the third (0.78), already read as such before correction; the confidence intervals carrying the other claims do not correct the same way and must be read as measurements, each with its own margin, not as independent rejections of a common null hypothesis [c7-multiplicite.md].

The full census covers 47 adjudicated tests across three families — the article’s claims, the standalone controls, and the abandoned branches — with 25 confirmed and 15 refuted. It was written

over the 15 c7-* sub-studies existing at the time and does not include the four verdicts added the same day (the strong attacker, DP, downstream utility, and the paid frontier-model run), which is why Table 1 and the census do not carry the same totals.

A second pass extended the confirmatory family to 15 tests convertible to a p-value and applied both Holm and Benjamini-Hochberg. **Neither changes any verdict**, and the two agree on all 15: the p-values are bimodal — eight at or below 0.0025, seven already near 1 — so no claim here rests on a marginally significant result [c7-multiplicite-globale-2026-09-12.md § 3]. It also flags three thresholds as fragile, which we name rather than let a reviewer find: the frontier twin’s **fidelity** > **0.10** bar, confirmed at 0.1714 but refuted had it been set at 0.18 — while our own weakest known twin sits at 0.177, above the value that cleared it; the choice of k in top-k (§ 5.2); and the functional form of the scale law beyond the measured

range (§ 5.9). A common bootstrap across analyses reusing the same 2,058 people was abandoned for cost, so the intervals of § 5.1–§ 5.3 on Twin are correct individually and carry no simultaneous coverage.

7.4 Scope, and two limits established by failure

Two datasets, one language, one questionnaire format, categorical closed-choice answers only. The mechanism analysis and the defense both rest on a single block of a single dataset. We make no claim about conversational agents or panels outside the two studied here. Two further limits we state with the specific resource that would lift them, because we looked for that resource and did not find it.

Generality beyond GSS and Big Five instruments. Both our datasets are American survey panels built on the same instrument family, so we cannot say whether this channel is a property of that family. We searched our holdings for a third dataset of a different nature carrying both real answers and simulated outputs for the same people, and found none usable [c7-troisieme-jeu-inventaire-2026-09-12.md]: the GSS panel is the very instrument a reviewer would object to; the Westwood PNAS 2025 material has no matched human file; NORC mode data and the ANES codebooks hold no individual microdata. Two candidates are genuinely different in nature and publishable in aggregate — the NY Fed Survey of Consumer Expectations (82,535 observations, stable user id) and Ahler-Sood (YouGov, CC0) — but **neither has existing simulated outputs** for its respondents. Generating them ourselves would introduce a new confound, our own generation method and models, and would test a different question. The missing resource is an individual dataset from a domain distinct from GSS and Big Five — health, consumption, elections — for which LLM twins produced by a **third-party team** already exist for the same respondents, under a licence permitting local computation and publication of an aggregate rate.

Free-text outputs. We measure nothing about free text, and the reason is that neither dataset permits it: the Park archive publishes **no transcripts**, and **none of the 13 Twin configurations** produces text. Every rate in this paper therefore concerns closed-choice categorical answers. The missing resource is Park’s complete transcripts, or a run of the Twin pipeline that generates text. We extrapolate our rates to free-text outputs in neither direction — note that the free-text inference literature [23, 35] attacks a richer signal than ours, so our figures are not an upper bound on that setting.

Ethical Considerations

Datasets, and the terms we are bound by. Twin-2K-500 is public under CC BY 4.0. The Park replication package is publicly posted on OSF (node t6g7k), but public posting is not a licence: the OSF API returns `node_license: null` for that node, so **no licence is declared** and public access carries no right to redistribute the individual responses it contains. Its questionnaire content is a General Social Survey instrument, and **NORC’s terms of use forbid reproducing the GSS in any form without prior written consent**. We record both constraints because an article about privacy that passes over the terms of use of its own data disqualifies itself.

What we do in consequence, verified rather than assumed. We redistribute no individual response file from either archive. We

checked our own released material file by file: the review artifact (§ 9) ships only code and a synthetic dataset generated at run time and reads nothing from `data/`, which is excluded from version control; and every published table under `resultats/` is an aggregate by condition, by item or by bootstrap replicate, carrying no individual record and no participant identifier. Only analysis code and disclosure-reviewed aggregates are published, with the NORC citation (Smith, Davern, Freese and Morgan, *General Social Surveys, panel files*, Chicago: NORC) and attribution to Park et al. One limit we state rather than resolve: NORC’s published terms address reproduction of the GSS and do not speak to derived aggregate statistics, and we found no primary source settling that case — the reading that per-item and per-condition aggregates fall outside the prohibition is ours, not NORC’s.

No new data was collected: the attack uses only already-published content (human answers and twin outputs). A limit to note: participants did not specifically consent to a re-identification test on the twins generated from their answers.

No individual is identified. No name or identifier linking a `pid/TWIN_ID` to a real person is published or listed. Only rates aggregated over the full cohorts are reported; no individual case is named or shown.

No new harm on Twin-2K-500. Twin-2K-500 already publishes, for each person, their identifiers and real answers beside their twin: matching is trivial without any attack. The result does not create new harm on this specific dataset; it documents a generic risk for anyone releasing “anonymised” twins without that key — a practice Twin-2K-500 does not follow.

The Park archive is a different case, and we make no such claim there. That argument rests on a key published beside the twins, and it does not transfer: on the Park archive no trivial alignment makes the matching free, the rates we measure are far higher (90.40 % closed-world, 60.17 % open-world at 1 % false accusations), and the objects we attack are the very ones whose individual release the authors treated as a privacy matter. We therefore do not argue that this paper creates no new harm there. What limits the harm is narrower and we state only that: we publish no individual record and no per-person result from that archive, only aggregate rates; and the disclosure below is addressed to its authors, whose own warning this work quantifies.

Benefit. Several teams publish individualised LLM twin outputs without any linkage risk analysis. This provides a first measured figure for that risk on public datasets and a reproducible method for testing it elsewhere *before* publication.

Responsible disclosure, and its actual state. Letters to the Twin-2K-500 authors (Toubia et al.) and to Park et al. are drafted, with a 30-day response window and an offer to share code and report in advance. The tone toward Park et al. is fixed: *your warning was accurate, here is its measured magnitude. As of this submission the letters have not been sent and the 30-day window has not opened*. Sending them is a decision of the responsible investigator; it is on the critical path and is to happen before any preprint posting or code release. We write this plainly rather than imply a consultation that has not taken place: filing this manuscript is itself a disclosure event, and the committee should know the window was still closed when it was filed.

The Park et al. case. Their supplementary material (Participant Consent) already warned participants that their information might be “inadvertently shared” and acknowledged that complete anonymity remains challenging. They named the risk, obtained an ethics agreement worked over more than six months, pseudonymised, restricted access to part of the material and planned a 25-year withdrawal. What is nevertheless public is the replication package we analyse, which carries the 1,052 participants’ individual responses together with the agent conditions’ responses (§ 4.1): the protection announced in the paper does not cover the objects this attack consumes. We flag that gap as a question for the authors rather than a finding against them — from outside we cannot tell whether the restriction was lifted or never covered the replication package — and it is precisely what the disclosure letter must state correctly. No public audit had measured the magnitude of that named risk; our figures (90.40 % for the composite agent among 1,052 in a closed world, 60.17 % in an open world at 1 % false accusations, and 44.7 % — a lower bound, the strong attack not re-run on the interview-only condition — from the interview alone) are, to our knowledge, its first quantification. We note for the disclosure letter that our own first measurement, 65.51 % closed-world and 20.39 % open-world, understated the magnitude of the risk they had named.

Publishing the attack code, for and against. *For:* replicability, and enabling other panel holders to test their twins before publication — standard security practice. *Against:* it lowers the cost for an attacker already holding a third-party panel’s real answers and a corresponding twin file. *Our resolution:* publish, because the code only acts if the attacker already holds real answers to the same items (rare outside the panel holder), and the countermeasure is straightforward once the risk is known.

Recommended defenses. (a) Do not publish twin outputs item by item for blocks with high inter-person specificity; publish aggregates or calibrated noise. (b) Break the person-twin alignment in public files. (c) Publish regenerated twins without copying wave metadata (StartDate/EndDate/Duration/RecordedDate). (d) Document a minimal linkage test before any twin release. (e) Within-segment shuffling (§ 6), with its real cost stated: 4.4 points on inter-item correlations, a 68 % worsening of the gap to human correlations (5.78 → 9.71), and the loss of every inter-item analysis downstream (§ 6.3) — it holds at 0.29 % against an attacker who knows the mechanism, but only for a split that shuffles the informative block (§ 6.1). (f) Differential privacy where the published object can be a population statistic rather than a per-person twin: it answers a different threat model, and buys its low leakage by never conditioning on an individual (§ 6.2).

Public interest. The AAPOR report of 8 May 2026 [32] ranks synthetic response generation as the most risky of the core tasks it evaluates and names re-identification by linkage without quantifying it. A risk named by practitioners and professional bodies but never measured justifies an independent measurement followed by responsible disclosure rather than silence.

Ethics review and scope. This study was not submitted to an external ethics panel such as an IRB. It relies exclusively on secondary data already made public by their original publishing teams — Twin-2K-500 (CC BY 4.0) and the Park et al. replication package — and no new data was collected from any person: we surveyed no one, interviewed no one, and had no contact with any

data subject. On that basis we judged the study out of scope for human-subjects review; § 8 above records the corresponding limit (participants did not specifically consent to a re-identification test on the twins generated from their answers).

This scoping does not extend to the released artifact, which processes no real person’s data at all (§ 9).

Open Science

Review artifact. `artefact/` replays the re-identification attack and its D4 defense in a single command (`./artefact/run.sh`), **with no real data and no real person**, in approximately 25 seconds on a laptop, without GPU, without network, and without any language model call. It runs on a fictional dataset of 600 synthetic persons generated at fixed seed (`config.GRAINE = 20260912`) with a tunable individual-signal dial (`config.SIGNAL_INDIVIDUEL = 0.25`), structurally comparable to the real data (categorical items, product × price matrix block, demographic segments, retest) but with different sample sizes and signal strength.

The artifact demonstrates that the two mechanics are the ones running in the paper: `attaque.py` imports `items_communs`, `rangs_attaque`, `rang_dans_segment` and `graine_nom` from `analyses/c7_reidentification.py`, and `defense.py` imports `defense_d4` and `mesurer_utilite` from `analyses/c7_defense.py`. Nothing is copied; the only function that reads real data (`t1_commun.charger()`) is never called, and a guard (`garde.py`) refuses to start if a path to `data/` is detected.

No artifact figure is a result of this study. The artifact’s outputs (on fictional data: 21.9 % before defense, 0.842 % after D4) illustrate the mechanics and must never be cited as study results. The study’s figures are those of § 5 and § 6, measured on Twin-2K-500 and the Park archive.

Obtaining the real data. Twin-2K-500: Hugging Face repository LLM-Digital-Twin/Twin-2K-500, CC BY 4.0. Park replication archive: publicly posted on OSF at <https://osf.io/t6g7k/>; it contains real individual responses from 1,052 human participants, is not redistributed here, and is never read by the artifact. Two conditions bear on obtaining it, and public availability satisfies neither: **no licence is declared** on that OSF node (`node_license: null`), so public access confers no redistribution right; and the GSS content it carries falls under **NORC’s terms of use, which forbid reproducing the GSS in any form without prior written consent** (§ 8). Obtain both datasets from their original sources and verify current terms of use there before any download.

Preregistrations. `resultats/c7-preenregistrement.md`, `resultats/c7-defense-preenregistrement.md`, `resultats/c7-stanford-preenregistrement.md`, each written before any computation. Third-party timestamping is pending.

Environment. Python 3.13.14, numpy 2.5.2, pandas 3.0.5, scikit-learn 1.9.0. No network dependency.

AI Use

Large-language-model-based agents were used throughout this project: they wrote the analysis code, ran the experiments, drafted this manuscript, and conducted adversarial review of both the code and the text. We state this in full rather than narrowly, because an

account that understated that involvement would mislead a reader more than a plain one.

The responsible investigator directed the work throughout: set the research questions and the falsifiable predictions in advance of each analysis, decided what to keep, discard, or redo, authorized all paid computation, and is responsible for the accuracy and integrity of the submission, including this section. No generative AI tool is listed as an author.

Three checks bear on how much trust the model-produced outputs warrant. First, quantitative predictions were preregistered in writing, dated, before the corresponding analysis script was run; the repository holds several dozen such preregistrations, one per analysis. Second, the analysis pipeline fixes a random seed at every stochastic step, and independent reruns are checked to reproduce identical output before a result is reported. Third, the manuscript and code were subjected to adversarial review passes whose default posture is rejection; these passes are how two substantive defects were found and corrected before submission: a statistical control meant to carry only each person's accuracy margin that in fact copied their true individual answer with the complementary probability, and a bootstrap confidence interval reported as "[0, 0]" on a zero-event sample — an artifact of the percentile method rather than a genuine null result, since replaced with an exact Clopper-Pearson interval.

These checks reduce, but do not remove, the risk that a model-produced number or claim is wrong; the responsible investigator remains accountable for every figure and statement in this paper regardless of how it was produced.

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